

EXEPERIMENT 7

#exp 7\src\App.css

```
body {  
    margin: 0;  
    font-family: Arial, sans-serif;  
}  
  
.header {  
    background: #2d7be5;  
    color: white;  
    text-align: center;  
    padding: 25px;  
    font-size: 25px;  
}  
  
.navbar {  
    background: #222;  
    color: white;  
    display: flex;  
    justify-content: space-between;  
    padding: 10px 20px;  
    font-size: 25px;  
}  
  
.navbar a {  
    color: white;  
    text-decoration: none;  
    margin-left: 15px;  
}  
  
.container {  
    display: flex;  
    padding: 40px;  
    background: #eee;  
}  
  
.column {  
    flex: 1;  
    padding: 20px;
```

```
}  
.footer {  
  background: #333;  
  color: white;  
  text-align: center;  
  padding: 15px;  
  position: fixed;  
  width: 100%;  
  bottom: 0;  
}
```

#exp 7\src\App.jsx

```
import { StrictMode } from 'react'  
import { createRoot } from 'react-dom/client'  
import './App.css'
```

```
function Student({ name, course, marks }) {  
  return (  
    <div className="student-card">  
      <h2>{name}</h2>  
      <p>Course: {course}</p>  
      <p>Marks: {marks}</p>  
    </div>  
  )  
}  
  
function App() {  
  return (  
    <div className="app-container">  
      <h1 className="app-title">Student Information</h1>  
      <Student name="Rahul Sharma" course="CSE" marks={80} />  
      <Student name="Anita Verma" course="IT" marks={92} />  
      <Student name="Rohan Gupta" course="Electronics" marks={78} />  
      <p>
```

NAME: PRIYANSHU KUMAR

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SECTION: CSE 26

</p>

</div>

)

}

export default App

#exp 7\src\index.css

body{

text-align: center;

margin: 0;

padding: 0;

}

#exp 7\src\main.jsx

import { StrictMode } from 'react'

import { createRoot } from 'react-dom/client'

import './index.css'

import App from './App.jsx'

createRoot(document.getElementById('root')).render(

<StrictMode>

<App />

</StrictMode>,

)

EXPERIMENT 8

#exp 8\src\App.css

```
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}  
  
body {  
  font-family: 'Inter', 'Segoe UI', sans-serif;  
  background: #0f172a;  
  min-height: 100vh;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  padding: 24px;  
  color: #ffffff;  
}  
  
.container {  
  width: 100%;  
  max-width: 720px;  
  min-height: 520px;  
  background: #1e293b;  
  border: 1px solid rgba(255,255,255,0.08);  
  border-radius: 8px;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  padding: 24px;  
  box-shadow: 0 10px 25px rgba(0,0,0,0.25);  
}  
  
.card {  
  width: 100%;  
  max-width: 460px;
```

```
background: #111827;
border: 1px solid #334155;
border-radius: 8px;
padding: 40px;
text-align: center;
box-shadow: 0 8px 20px rgba(0,0,0,0.2);
}
```

```
h1 {
  font-size: 36px;
  font-weight: 700;
  margin-bottom: 24px;
  color: #ffffff;
}
```

```
h2 {
  font-size: 72px;
  font-weight: 700;
  color: #38bdf8;
  margin: 24px 0;
}
```

```
.button-container {
  display: flex;
  justify-content: center;
  gap: 16px;
  flex-wrap: wrap;
  margin-top: 24px;
}
```

```
button {
  background: #38bdf8;
  color: #082f49;
  border: none;
  padding: 14px 22px;
  border-radius: 8px;
  font-size: 15px;
  font-weight: 700;
```

```
    cursor: pointer;
}

.reset-btn {
    margin-top: 20px;
    background: #ef4444;
    color: white;
}

@media(max-width: 768px) {
    .container {
        min-height: auto;
        padding: 20px;
    }
    .card {
        padding: 28px;
    }
    h1 {
        font-size: 28px;
    }
    h2 {
        font-size: 56px;
    }
    .button-container {
        flex-direction: column;
    }
    button {
        width: 100%;
    }
}
```

```
import { useState } from "react";
import "./App.css";
function App() {
  const [count, setCount] = useState(0);
  return (
    <div className="container">
      <div className="card">
        <h1>React Counter Application</h1>
        <h2>{count}</h2>
        <div className="button-container">
          <button onClick={() => setCount(count + 1)}>
            Increment (+)
          </button>
          <button onClick={() => setCount(count - 1)}>
            Decrement (-)
          </button>
        </div>
        <button
          className="reset-btn"
          onClick={() => setCount(0)}
        >
          Reset
        </button>
        <br />
        <br />
        <div><p>NAME: PRIYANSHU KUMAR<br />
          ROLL.NO: 2503201000871 <br />
          SECTION: CSE 26 <br />
        </p></div>
      </div>
    </div>
  );
}
```

```
}  
export default App;
```

#exp 8\src\main.jsx

```
import { StrictMode } from 'react'  
import { createRoot } from 'react-dom/client'  
import './index.css'  
import App from './App.jsx'  
  
createRoot(document.getElementById('root')).render(  
  <StrictMode>  
    <App />  
  </StrictMode>,  
)
```


EXEPERIMENT 9

#exp 9\src\App.css

```
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}  
  
body {  
  font-family: 'Inter', 'Segoe UI', sans-serif;  
  background: #0f172a;  
  min-height: 100vh;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  padding: 24px;  
  color: #fff;  
}  
  
.container {  
  width: 100%;  
  max-width: 460px;  
  padding: 36px;  
  background: #1e293b;  
  border: 1px solid rgba(255,255,255,0.08);  
  border-radius: 8px;  
  box-shadow: 0 10px 25px rgba(0,0,0,0.25);  
}  
  
.container h1 {  
  text-align: center;  
  font-size: 30px;  
  font-weight: 700;  
  margin-bottom: 28px;
```

```
    color: #ffffff;
}

form {
    display: flex;
    flex-direction: column;
    gap: 6px;
}

label {
    font-size: 13px;
    font-weight: 600;
    color: #cbd5e1;
    margin-top: 12px;
    margin-bottom: 6px;
}

input {
    width: 100%;
    padding: 14px 16px;
    font-size: 15px;
    color: #fff;
    background: #0f172a;
    border: 1px solid #334155;
    border-radius: 8px;
    outline: none;
}

input::placeholder {
    color: #94a3b8;
}

input:focus {
    border-color: #38bdf8;
}
```

```
.error {  
    font-size: 12px;  
    color: #f87171;  
    margin-top: 4px;  
}  
  
button[type="submit"] {  
    margin-top: 24px;  
    padding: 14px;  
    font-size: 15px;  
    font-weight: 700;  
    color: #082f49;  
    background: #38bdf8;  
    border: none;  
    border-radius: 8px;  
    cursor: pointer;  
}  
  
.apiData {  
    margin-top: 28px;  
    padding-top: 12px;  
    border-top: 1px solid #334155;  
}  
  
.apiData h2 {  
    font-size: 18px;  
    text-align: center;  
    margin-bottom: 18px;  
    color: #ffffff;  
}  
  
.apiData p {  
    background: #0f172a;  
    border-left: 4px solid #38bdf8;  
    border-radius: 8px;  
    padding: 16px;
```

```
font-size: 15px;
color: #e2e8f0;
line-height: 2;
}
```

```
.user-label {
  font-weight: 700;
  color: #38bdf8;
}
```

#exp 9\src\App.jsx

```
import { useState } from 'react'
import './App.css'
```

```
function App(){
  const [name, setName] = useState("")
  const [email, setEmail] = useState("")
  const [password, setPassword] = useState("")
  const [errors, setErrors] = useState({})
  const [registeredUser, setRegisteredUser] = useState(null)

  const handleSubmit = (e) => {
    e.preventDefault();
    let formErrors = {};

    if (name.trim() === "") {
      formErrors.name = "Name is required"
    }

    if (!email.includes("@")) {
      formErrors.email = "Enter a valid email address"
    }

    if (password.length < 6) {
      formErrors.password = "Password must be at least 6 characters"
    }
  }
}
```

```
setErrors(formErrors);
```

```
if (Object.keys(formErrors).length === 0) {
```

```
  setRegisteredUser({ name, email });
```

```
  setName("");
```

```
  setEmail("");
```

```
  setPassword("");
```

```
  setErrors({});
```

```
}
```

```
}
```

```
return (
```

```
<div className="container">
```

```
<h1>Registration Form</h1>
```

```
<form onSubmit={handleSubmit}>
```

```
<label>Name</label>
```

```
<input
```

```
  type="text"
```

```
  placeholder="Enter your name"
```

```
  value={name}
```

```
  onChange={(e) => setName(e.target.value)}
```

```
/>
```

```
{errors.name && <p className="error">{errors.name}</p>}
```

```
<label>Email</label>
```

```
<input
```

```
  type="email"
```

```
  placeholder="Enter your email"
```

```
  value={email}
```

```
  onChange={(e) => setEmail(e.target.value)}
```

```
/>
```

```
{errors.email && <p className="error">{errors.email}</p>}
```

```

<label>Password</label>

<input
  type="password"
  placeholder="Enter your password"
  value={password}
  onChange={(e) => setPassword(e.target.value)}
/>

{errors.password && <p className="error">{errors.password}</p>}

<button type="submit">Register</button>
</form>

{registeredUser && (
  <div className="apiData">
    <h2>Registered User</h2>
    <p>
      <span className="user-label">Name:</span> {registeredUser.name}<br/>
      <span className="user-label">Email:</span> {registeredUser.email}
    </p>
  </div>
)} <br />
<div className="container">
  <p>
    NAME: PRIYANSHU KUMAR <br />
    ROLL.NO: 2503201000871 <br />
    SECTION: CSE 26 <br />
  </p>
</div>
</div>

);
}

export default App;

```

#exp 9\src\index.css

```
* {  
  box-sizing: border-box;  
  margin: 0;  
  padding: 0;  
}  
  
body {  
  background-color: #f0f2f5;  
  font-family: 'Segoe UI', Roboto, sans-serif;  
  min-height: 100vh;  
}  
  
#root {  
  min-height: 100vh;  
  display: flex;  
  flex-direction: column;  
}
```

#exp 9\src\main.jsx

```
import { StrictMode } from 'react'  
import { createRoot } from 'react-dom/client'  
import './index.css'  
import App from './App.jsx'  
  
createRoot(document.getElementById('root')).render(  
  <StrictMode>  
    <App />  
  </StrictMode>,  
)
```

EXEPERIMENT 1

#exp 1\1.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Document</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <h1 class="text-center text-primary">My Website</h1>

  <nav class="navbar navbar-expand-lg navbar-light bg-dark navbar-dark">

    <div class="container-fluid">

      <a class="navbar-brand" href="#">My Website</a>

      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav"
        aria-controls="navbarNav" aria-expanded="false" aria-label="Toggle navigation">

        <span class="navbar-toggler-icon"></span>

      </button>

      <div class="collapse navbar-collapse" id="navbarNav">

        <ul class="navbar-nav">

          <li class="nav-item">

            <a class="nav-link active" aria-current="page" href="#">Home</a>

          </li>

          <li class="nav-item">

            <a class="nav-link" href="#">About</a>

          </li>

          <li class="nav-item">
```



```

        <a class="nav-link" href="#">Contact</a>
    </li>
</ul>
</div>
</div>
</nav>
<div class="container">
    <div class="column">
        <h2>Column 1</h2>
        <p>This is the first column. On small devices the column occupies full width, while on
            medium and large devices it occupies half width using Bootstrap grid classes.</p>
    </div>
    <div style="border-left: 3px solid black; height: 200px;"></div>
    <div class="column">
        <h2>Column 2</h2>
        <p>This is the second column. Bootstrap grid system helps to create responsive
            layout.</p>
    </div>
</div>
<div class="footer">
    NAME: PRIYANSHU KUMAR <br>ROLL.NO: 2503201000871 <br>SECTION: CSE 26
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>

</html>

```

#exp 1\style.css

```

body {
    margin: 0;
    font-family: Arial, sans-serif;
}

```

```

.header {

```

```
background: #2d7be5;
color: white;
text-align: center;
padding: 25px;
font-size: 25px;
}
```

```
.navbar {
background: #222;
color: white;
display: flex;
justify-content: space-between;
padding: 10px 20px;
font-size: 25px;
}
```

```
.navbar a {
color: white;
text-decoration: none;
margin-left: 15px;
}
```

```
.container {
display: flex;
padding: 40px;
background: #eee;
}
```

```
.column {
flex: 1;
padding: 20px;
}
```

```
.footer {
```

```
background: #333;  
color: white;  
text-align: center;  
padding: 15px;  
position: fixed;  
width: 100%;  
bottom: 0;  
}
```

EXEPERIMENT 2

```
<!DOCTYPE html>

<html>

<head>

  <title>TechStore</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
      <a class="navbar-brand" href="#">TechStore</a>
      <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#menu">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="menu">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item"><a class="nav-link" href="#">Home</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Products</a></li>
          <li class="nav-item"><a class="nav-link" href="#">About</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
        </ul>
      </div>
    </div>
  </nav>

  <div class="container mt-5">
    <div class="row">

      <div class="col-lg-4 col-md-6 col-12 mb-4">
        <div class="card">
```

```

<div class="card-body">
  <h5>Laptop</h5>
  <p>High performance laptop</p>
  <h6>₹50,000 <span class="badge bg-success">New</span></h6>
  <button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#m1">View
    Details</button>
</div>
</div>
</div>
<div class="col-lg-4 col-md-6 col-12 mb-4">
  <div class="card">
    
    <div class="card-body">
      <h5>Mobile</h5>
      <p>Latest smartphone</p>
      <h6>₹30,000 <span class="badge bg-danger">Sale</span></h6>
      <button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#m2">View
        Details</button>
    </div>
  </div>
</div>
</div>
<div class="col-lg-4 col-md-6 col-12 mb-4">
  <div class="card">
    
    <div class="card-body">
      <h5>Headphones</h5>
      <p>Noise cancelling</p>
      <h6>₹5,000 <span class="badge bg-success">New</span></h6>
      <button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#m3">View
```

```
        Details</button>
    </div>
</div>
</div>

</div>
</div>
<div class="modal fade" id="m1">
    <div class="modal-dialog">
        <div class="modal-content">
            <div class="modal-header">
                <h5>Laptop Details</h5>
                <button class="btn-close" data-bs-dismiss="modal"></button>
            </div>
            <div class="modal-body">
                16GB RAM, SSD, Fast performance
            </div>
            <div class="modal-footer">
                <button class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
            </div>
        </div>
    </div>
</div>
</div>

<div class="modal fade" id="m2">
    <div class="modal-dialog">
        <div class="modal-content">
            <div class="modal-header">
                <h5>Mobile Details</h5>
                <button class="btn-close" data-bs-dismiss="modal"></button>
            </div>
            <div class="modal-body">
                High camera quality smartphone
            </div>
```

```

        <div class="modal-footer">
            <button class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
        </div>
    </div>
</div>
</div>

<div class="modal fade" id="m3">
    <div class="modal-dialog">
        <div class="modal-content">
            <div class="modal-header">
                <h5>Headphones Details</h5>
                <button class="btn-close" data-bs-dismiss="modal"></button>
            </div>
            <div class="modal-body">
                Best sound quality with bass
            </div>
            <div class="modal-footer">
                <button class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
            </div>
        </div>
    </div>
</div>

<footer class="bg-dark text-white text-center p-3">
    <p>NAME: PRIYANSHU KUMAR <br>ROLL.NO: 2503201000871 <br>SECTION: CSE 26</p>
</footer>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>
</body>
</html>

```

```
body{  
background:#ffffff;  
}
```

```
.card:hover{  
transform: scale(1.05);  
transition: 0.3s;  
}
```

```
.card{  
background-color: rgb(0, 0, 0);  
color: rgb(255, 255, 255);  
text-align: center;  
}
```

```
footer{  
margin-top:50px;  
position: relative;  
bottom: 0;  
width: 100%;  
}
```


EXEPERIMENT 3

#exp 3\exp3.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Health Care Plus</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-gray-200 font-sans">
  <header class="bg-gradient-to-r from-slate-600 to-indigo-400 text-white p-4">
    <div class="flex justify-between items-center">
      <h1 class="text-xl font-bold">Health Care Plus</h1>
      <nav class="space-x-4 text-sm">
        <a href="#" class="hover:text-gray-200">Home</a>
        <a href="#" class="hover:text-gray-200">Programs</a>
        <a href="#" class="hover:text-gray-200">Trainers</a>
        <a href="#" class="hover:text-gray-200">Contact</a>
      </nav>
    </div>
  </header>
  <section class="bg-gradient-to-r from-sky-200 to-indigo-400 text-white text-center py-16">
    <h2 class="text-4xl font-bold mb-4">
      Transform Your Body & Mind
    </h2>
    <p class="mb-6 text-lg">
      Join Health Care Plus today and achieve your health goals with expert trainers.
    </p>
    <button class="bg-gray-200 text-gray-700 px-6 py-2 rounded-md font-semibold hover:bg
white">
```

Join Now

</button> </section>

<section class="py-12 px-6">

<h3 class="text-3xl font-bold text-center mb-10">

Our Programs

</h3>

<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">

<div class="bg-gray-100 p-6 rounded-lg shadow text-center">

<h4 class="text-red-600 font-bold mb-2">

Weight Training </h4>

<p>

Build strength and muscle with guided weight training programs.

</p></div>

<div class="bg-gray-100 p-6 rounded-lg shadow text-center">

<h4 class="text-red-600 font-bold mb-2">

Yoga Classes

</h4>

<p>

Improve flexibility and mental wellness through yoga sessions.

</p> </div>

<div class="bg-gray-100 p-6 rounded-lg shadow text-center">

<h4 class="text-red-600 font-bold mb-2">

Diet Plans

</h4>

<p>

Personalized diet plans to complement your fitness routine.

</p></div></div>

</section>

<footer class="bg-gradient-to-r from-sky-300 to-indigo-400 text-black text-center py-6">

<p>NAME: PRIYANSHU KUMAR
ROLL.NO: 2503201000871
SECTION: CSE 26</p>

</footer>

</body>

</html>

EXEPERIMENT 10

```
// Abstract class abstract class BankAccount {    private int accountNumber;    private String accountHolderName;
private double balance;
```

```
// Constructor
```

```
    BankAccount(int accountNumber, String accountHolderName, double balance) {        this.accountNumber =
accountNumber;        this.accountHolderName = accountHolderName;        this.balance = balance;
    }
```

```
// Getters    public int getAccountNumber() {        return accountNumber;
    }
```

```
    public String getAccountHolderName() {        return accountHolderName;
    }
```

```
    public double getBalance() {        return balance;
    }
```

```
// Setter
```

```
    public void setBalance(double balance) {        this.balance = balance;
    }
```

```
// Deposit Method    void deposit(double amount) {        balance += amount;
        System.out.println("Amount Deposited: " + amount);
    }
```

```
// Display Method    void displayDetails() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder Name: " + accountHolderName);
        System.out.println("Balance: " + balance);
    }
```

```
// Abstract Method    abstract void calculateInterest();
}
```

```
// SavingsAccount

class SavingsAccount extends BankAccount {

    SavingsAccount(int accountNumber, String accountHolderName, double balance) {        super(accountNumber,
accountHolderName, balance);

    }

    void calculateInterest() {

        double interest = getBalance() * 0.05;

        System.out.println("Savings Account Interest: " + interest);

    }

}
```

```
// CurrentAccount

class CurrentAccount extends BankAccount {

    CurrentAccount(int accountNumber, String accountHolderName, double balance) {        super(accountNumber,
accountHolderName, balance);

    }

    void calculateInterest() {

        double interest = getBalance() * 0.02;

        System.out.println("Current Account Interest: " + interest);

    }

}
```

```
// Main Class public class Exp10 {    public static void main(String[] args) {

    SavingsAccount sa = new SavingsAccount(101, "Rahul", 10000);

    CurrentAccount ca = new CurrentAccount(102, "Anita", 20000);

    System.out.println("----- Savings Account -----");    sa.deposit(2000);    sa.displayDetails();
sa.calculateInterest();

    System.out.println();

}
```

```
        System.out.println("----- Current Account -----");    ca.deposit(3000);    ca.displayDetails();
ca.calculateInterest();
    }
}
```

Output - Exp10 (run) ×

```
----- Savings Account -----
Amount Deposited: 2000.0
Account Number: 101
Account Holder Name: Rahul
Balance: 12000.0
Savings Account Interest: 600.0

----- Current Account -----
Amount Deposited: 3000.0
Account Number: 102
Account Holder Name: Anita
Balance: 23000.0
Current Account Interest: 460.0
```

EXEPERIMENT 11

```
<project xmlns="http://maven.apache.org/POM/4.0.0"      xmlns:xsi="http://www.w3.org/2001/XMLSchema-  
instance"      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
https://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.example</groupId>
```

```
<artifactId>student-rest-api</artifactId>
```

```
<version>0.0.1-SNAPSHOT</version>
```

```
<parent>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-parent</artifactId>
```

```
    <version>3.2.5</version>
```

```
</parent>
```

```
<properties>
```

```
    <java.version>17</java.version>
```

```
</properties>
```

```
<dependencies>
```

```
    <!-- Spring Web Dependency -->
```

```
    <dependency>
```

```
        <groupId>org.springframework.boot</groupId>
```

```
        <artifactId>spring-boot-starter-web</artifactId>
```

```
    </dependency>
```

```
</dependencies>
```

```
<build>
```

```
    <plugins>
```

```
        <plugin>
```

```
            <groupId>org.springframework.boot</groupId>
```

```
            <artifactId>spring-boot-maven-plugin</artifactId>
```

```
</plugin>
</plugins>
</build>

</project>
```

```
package com.example.student;
```

```
// Model class representing Student entity public class Student {
```

```
    private int id;    private String name;    private String course;
```

```
    // Default constructor    public Student() {
    }
```

```
    // Parameterized constructor    public Student(int id, String name, String course) {
        this.id = id;
        this.name = name;    this.course = course;
    }
```

```
    // Getter and Setter methods    public int getId() {    return id;
    }
```

```
    public void setId(int id) {    this.id = id;
    }
```

```
    public String getName() {    return name;
    }
```

```
    public void setName(String name) {    this.name = name;
    }
```

```
    public String getCourse() {    return course;
```

```
    }  
    public void setCourse(String course) {        this.course = course;  
    }  
}
```

```
package com.example.student;
```

```
import org.springframework.http.HttpStatus; import org.springframework.http.ResponseEntity; import  
org.springframework.web.bind.annotation.*;
```

```
// REST Controller
```

```
@RestController
```

```
@RequestMapping("/") public class StudentController {
```

```
    private Student student;
```

```
    // Welcome endpoint    @GetMapping    public String welcomeMessage() {        return "Welcome to Student  
Management REST API";  
    }
```

```
    // GET student details    @GetMapping("/student")    public ResponseEntity<Student> getStudent() {        if  
(student == null) {  
        return new ResponseEntity<>(HttpStatus.NOT_FOUND);  
    }
```

```
        return new ResponseEntity<>(student, HttpStatus.OK);  
    }
```

```
    // POST student data    @PostMapping("/student")    public ResponseEntity<Student>  
createStudent(@RequestBody Student student) {
```

```
        this.student = student;        return new ResponseEntity<>(student, HttpStatus.CREATED);  
    }
```

```
}
```



```
package com.example.student;
```

```
import org.springframework.boot.SpringApplication; import  
org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
// Main Spring Boot Application @SpringBootApplication public class StudentRestApiApplication {
```

```
    public static void main(String[] args) {        SpringApplication.run(StudentRestApiApplication.class, args);  
    }  
}
```

EXEPERIMENT 12

```
<project xmlns="http://maven.apache.org/POM/4.0.0"      xmlns:xsi="http://www.w3.org/2001/XMLSchema-  
instance"      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
https://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
  <modelVersion>4.0.0</modelVersion>  <groupId>com.example</groupId>
```

```
  <artifactId>employeeapi</artifactId>
```

```
  <version>0.0.1-SNAPSHOT</version>
```

```
  <name>employeeapi</name>
```

```
  <parent>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-parent</artifactId>
```

```
    <version>3.2.5</version>
```

```
  </parent>
```

```
  <properties>
```

```
    <java.version>17</java.version>
```

```
  </properties>
```

```
  <dependencies>
```

```
    <!-- Spring Web -->
```

```
    <dependency>
```

```
      <groupId>org.springframework.boot</groupId>
```

```
      <artifactId>spring-boot-starter-web</artifactId>
```

```
    </dependency>
```

```
    <!-- Spring Data JPA -->
```

```
    <dependency>
```

```
      <groupId>org.springframework.boot</groupId>
```

```
      <artifactId>spring-boot-starter-data-jpa</artifactId>
```

```
    </dependency>
```

```
    <!-- H2 Database -->
```

```

<dependency>
  <groupId>com.h2database</groupId>
  <artifactId>h2</artifactId>
  <scope>runtime</scope>
</dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
</project>

```

Main class package com.example.employeeapi;

```

import org.springframework.boot.SpringApplication; import
org.springframework.boot.autoconfigure.SpringBootApplication;

```

```

@SpringBootApplication public class EmployeeApiApplication {   public static void main(String[] args) {
SpringApplication.run(EmployeeApiApplication.class, args);

  }
}

```

Employee class package com.example.employeeapi.entity;

```

import jakarta.persistence.Entity; import jakarta.persistence.Id;

```

```

@Entity public class Employee {

```

```

  @Id   private int id;   private String name;   private String department;   private double salary;

```

```
public Employee() {  
    }  
  
public Employee(int id, String name, String department, double salary) {    this.id = id;  
    this.name = name;    this.department = department;    this.salary = salary;  
    }  
  
public int getId() {    return id;  
    }  
  
public void setId(int id) {    this.id = id;  
    }  
  
public String getName() {    return name;  
    }  
  
public void setName(String name) {    this.name = name;  
    }  
  
public String getDepartment() {    return department;  
    }  
  
public void setDepartment(String department) {    this.department = department;  
    }  
  
public double getSalary() {    return salary;  
    }  
  
public void setSalary(double salary) {    this.salary = salary;  
    }  
}
```

```
package com.example.employeeapi.repository;
```

```
import org.springframework.data.jpa.repository.JpaRepository; import com.example.employeeapi.entity.Employee;
```

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
}
```

```
Employee service package com.example.employeeapi.service;
```

```
import java.util.List; import org.springframework.beans.factory.annotation.Autowired; import  
org.springframework.stereotype.Service; import com.example.employeeapi.entity.Employee; import  
com.example.employeeapi.repository.EmployeeRepository;
```

```
@Service public class EmployeeService {    @Autowired    private EmployeeRepository repository;
```

```
    // Add Employee    public Employee saveEmployee(Employee employee) {        return repository.save(employee);  
    }
```

```
    // Get All Employees    public List<Employee> getAllEmployees() {        return repository.findAll();  
    }
```

```
    // Get Employee By ID    public Employee getEmployeeById(int id) {        return repository.findById(id).orElse(null);  
    }
```

```
    // Update Employee
```

```
    public Employee updateEmployee(int id, Employee employeeDetails) {
```

```
        Employee employee = repository.findById(id).orElse(null);
```

```
        if (employee != null) {            employee.setName(employeeDetails.getName());  
employee.setDepartment(employeeDetails.getDepartment());  
employee.setSalary(employeeDetails.getSalary());            return repository.save(employee);  
        }  
        return null;  
    }
```

```

    // Delete Employee    public String deleteEmployee(int id) {    repository.deleteById(id);    return "Employee
deleted successfully!";

    }

}

```

Employee controller package com.example.employeeapi.controller;

```

import java.util.List; import org.springframework.beans.factory.annotation.Autowired; import
org.springframework.web.bind.annotation.*;

```

```

import com.example.employeeapi.entity.Employee; import com.example.employeeapi.service.EmployeeService;

```

```

@RestController

```

```

@RequestMapping("/employees") public class EmployeeController {

```

```

    @Autowired    private EmployeeService service;    // POST: Add Employee    @PostMapping    public Employee
addEmployee(@RequestBody Employee employee) {    return service.saveEmployee(employee);

    }

```

```

    // GET: All Employees    @GetMapping    public List<Employee> getAllEmployees() {    return
service.getAllEmployees();

    }

```

```

    // GET: Employee By ID    @GetMapping("/{id}")    public Employee getEmployeeById(@PathVariable int id) {
return service.getEmployeeById(id);

    }

```

```

    // PUT: Update Employee    @PutMapping("/{id}")    public Employee updateEmployee(@PathVariable int id,
@RequestBody Employee employee) {    return service.updateEmployee(id, employee);

    }

```

```

    // DELETE: Delete Employee    @DeleteMapping("/{id}")    public String deleteEmployee(@PathVariable int id) {
return service.deleteEmployee(id); } }

```